IBM Quantum Composer Circuits

Student Lab Handout

Session Overview

Build and run 3 key circuits using IBM Quantum Composer. Work in **pairs**!

Session Objectives:

- Independently build/run 1–2 qubit circuits
- Observe superposition + entanglement histograms
- Download/organize results for lab notebooks
- Troubleshoot queues from M1 setup

Timed Agenda (75 minutes)

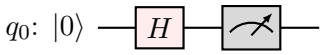
1. **0:00-0:10** Login check + M1 recap
2. **0:10-0:50** Paired circuit building (40 min)
3. **0:50-1:08** Share-out + discussion
4. **1:08-1:15** Collect + Week 3 preview

1 GET STARTED: quantum.ibm.com/composer

Pairs: One builds, one screenshots. **Switch roles** after each circuit.

2 CIRCUIT 1: Single-Qubit Superposition (15 min)

Build this:



Pink gate (H) Grey measure (M)

Expected: $|0\rangle$ ≈50%, $|1\rangle$ ≈50% (superposition)

Circuit 1 Instructions:

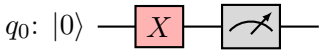
1. **Backend:** 'ibm_nairobi' (simulator) | **Shots:** '1024'
2. Drag H-gate (pink) to q_0
3. Drag Measure (grey) to end
4. **Run** → Screenshot histogram
5. **Save:** "Week2_Circuit1_Hgate"

Record your results:

$\|0\rangle$ counts	$\|1\rangle$ counts	Job ID	✓
_____	_____	_____	☐

3 CIRCUIT 2: Pauli X Flip (10 min)

Build this:



Red gate (X)

Expected: $|1\rangle = 100\%$ (deterministic flip)

Circuit 2 Instructions:

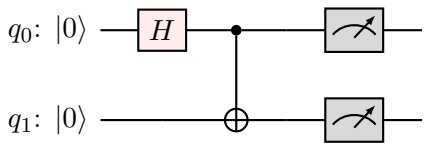
1. **Clear** Circuit 1 → New circuit
2. Drag X-gate (red) to q_0
3. Drag **Measure** to end
4. **Run** → Screenshot
5. **Save:** "Week2_Circuit2_Xgate"

Record your results:

$\|0\rangle$ counts	$\|1\rangle$ counts	Job ID	✓
_____	_____	_____	☐

4 CIRCUIT 3: 2-Qubit Bell State (20 min)

Build this:



H gate (pink) CNOT (solid •, open ◯)

Expected: $|00\rangle$ ≈50%, $|11\rangle$ ≈50% (ENTANGLEMENT!)

Circuit 3 Instructions:

1. Add **2 qubits** (drag from left)
2. q_0**:** H-gate → solid dot (control)
3. q_1**:** CNOT target (open circle + line)
4. Measure **both** qubits
5. **Run** → Screenshot
6. **Save:** "Week2_Circuit3_Bell"

Record your results:

$\|00\rangle$	$\|01\rangle$	$\|10\rangle$	$\|11\rangle$	Job ID	✓
_____	_____	_____	_____	_____	☐

5 SIMULATION REFERENCE (Expected Results)

```python
import numpy as np
shots = 1024

print("=== Circuit 1: H-gate ===")
print("|0>: 50%, |1>: 50%")

print("\n=== Circuit 2: X-gate ===")
print("|0>: 0%, |1>: 100%")

print("\n=== Circuit 3: Bell State ===")
print("|00>: 50%, |11>: 50%")
print("|01>, |10> should be ~0% (entanglement!)")
```

6 SUBMISSION TEMPLATE

Fill out + screenshot for credit:

Circuit 1 Results: $|0\rangle$=_ , $|1\rangle$=_ Job ID: _

Circuit 2 Results: $|0\rangle$=_ , $|1\rangle$=_ Job ID: _

Circuit 3 Results: $|00\rangle$=_ , $|11\rangle$=_ Job ID: _

Key Question: Why only $|00\rangle$ + $|11\rangle$? (Entanglement!) [web:22]

7 ✓COMPLETION CHECKLIST

All 3 circuits built + run (1024 shots)

Screenshots of all histograms

Job IDs recorded

Circuits saved (Week2_Circuit1/2/3)

Results table filled above

Submit notebook screenshots by week's end

8 DISCUSSION QUESTIONS

- Why do histograms vary between runs? (→ shots, simulation noise)
- Circuit 3: Why no $|01\rangle$ or $|10\rangle$ states? (→ entanglement)
- How many shots improve statistics?